



Department of Electronics and Communication Engineering Khulna University of Engineering & Technology

Proposal of Analog Electronics II Laboratory (ECE 2102)

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Group: B-2	
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2. **Title: Design and Implementation of a Line Follower Robot (LFR) Using Operational Amplifiers**

3. **Background and Motivation:**

The increasing demand for automation in industrial and domestic applications has made robotic systems an important area of research and development. Among different autonomous robots, the Line Follower Robot (LFR) is one of the most fundamental and widely used systems due to its simplicity, reliability, and practical applications in automated transportation, warehouse management, and industrial production lines. A line follower robot is designed to detect and follow a predefined path using sensors and control circuits.

The motivation behind this project is to design and implement a cost-effective and efficient LFR using operational amplifiers instead of complex microcontroller-based systems. Operational amplifiers provide fast analog signal processing, simple circuit design, and low power consumption, making them suitable for educational and beginner-level robotic applications. This project will help in understanding the practical implementation of sensors, comparators, motor driving mechanisms, and control systems. Furthermore, the project enhances knowledge in electronics, automation, and embedded system concepts while demonstrating how analog circuits can be effectively used to achieve autonomous robotic movement.

4. **Objectives:**

- To design and implement a Line Follower Robot (LFR) using operational amplifiers.
- To understand the working principle of line detection using IR sensors.
- To develop a control circuit for autonomous path following.
- To use operational amplifiers as comparators for signal processing.
- To control the movement of DC motors according to sensor outputs.
- To study the practical application of analog electronics in robotics.
- To achieve accurate and stable line tracking performance.
- To gain hands-on experience in circuit design, testing, and troubleshooting.

5. Conflicting Requirements:

- High speed vs accurate line tracking.
- Low cost vs high performance components.
- Low power consumption vs strong motor operation.
- Simple analog circuit design vs stable control system.
- Small robot size vs easy hardware arrangement.
- Fast sensor response vs noise reduction.
- Lightweight structure vs mechanical strength and durability.

6. Proposed Budget for Implementing the Project:

Sl.	Item	Justification	Price (BDT)
1.	Wheel	Used for the movement of the robot	320
2.	Battery	Provides power supply to the circuit and motor	180
3.	TP4056 Charger Module	Used for safe charging of the battery	80
4.	Battery Socket	Used for easy battery connection and removal	80
5.	Motor	Provides motion to the robot	600
6.	10 Ohm ,47 Ohm ,1k Ohm Resistor	Used for biasing ,voltage and current limiting	90
7.	LM358 IC	Used as an operational amplifier/comparator for signal processing	200
8.	Breadboard	Used for circuit prototyping and testing	160
9.	Ball Caster	Used for balance and smooth movement of the robot	200
10.	LDR Sensor	Used to detect light intensity	60
11.	10k Potentiometer	Used to adjust sensitivity/voltage level	200
12.	100 μ F Capacitor	Used for filtering and noise reduction	100
13.	LED	Used for indication and output display	40
14.	BC547 Transistor	Used for switching and signal amplification	100
17.	Wires	Used for connecting different components	200
Total (BDT)			2610
In words: Two thousand six hundred ten taka only			

7. Work Plan (Week wise plan):

7. Work Plan

1 st Week	Idea generating & problem solving
2 nd Week	Proteus implementation
3 rd Week	Project implementation and testing

8. Impact of project on the society, environment and sustainability:

The Line Follower Robot (LFR) project has positive impacts on society, industry, and education by promoting automation technology and improving efficiency in transportation and material handling systems. It helps reduce human effort in repetitive tasks and increases accuracy in industrial operations. The project also provides practical knowledge and skill development in robotics, electronics, and automation for students and researchers.

From an environmental perspective, the robot consumes low electrical power and supports energy-efficient operation. The use of compact electronic components minimizes material wastage and promotes eco-friendly system design.

In terms of sustainability, the project encourages the development of low-cost and reusable automation systems that can be applied in industries, hospitals, and smart transportation systems. The use of operational amplifiers instead of complex processors also reduces system complexity and maintenance cost, making the project more sustainable and economically beneficial for long-term applications.

Signature and Name of the Supervisor